

SHIVAM KUMAR MISHRA

NEW DELHI 110059

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Maharaja Agrasen Institute of Technology, GGSIPU

Aug 2023 – Present

Bachelor of Technology in Computer Science and Engineering

CGPA: 9.1 — New Delhi, India

Kendriya Vidyalaya, Sector-12 Dwarka

CBSE Board

Class XII: 94% — Class X: 93% — New Delhi, India

Relevant Coursework

- **Artificial Intelligence & Machine Learning:** Machine Learning, Deep Learning (TensorFlow, PyTorch), Reinforcement Learning, Computer Vision, Pattern Recognition, Natural Language Processing, Generative Modeling (VAEs, Diffusion Models)
- **Data Science & Analytics:** Data Mining, Big Data Analytics, Statistical Learning, Predictive Modeling, Applied Machine Learning
- **Systems & Security:** Information Security & Cryptography, Secure Communications (DRDO Internship), Backend Security Architecture (IWAI Internship), Vulnerability Assessment, Anomaly Detection, System Log Analysis
- **Mathematics & Foundations:** Linear Algebra, Probability & Statistics, Discrete Mathematics, Optimization Techniques, Numerical Methods
- **Advanced Topics:** Quantum Computing (CDAC Hyderabad, IIT Roorkee), Computational Materials Science, Responsible AI Governance

Experience

DESIDOC, DRDO (Ministry of Defence, Government of India)

Jan 2025 – Mar 2025

Cybersecurity & Machine Learning Intern

New Delhi, India

- Engineered a supervised learning–based email anomaly detection system, leveraging feature engineering and PyTorch pipelines to reduce anomalous emails by **60%** and improve spam detection accuracy by **25%**.
- Executed Vulnerability Assessment and Penetration Testing (VAPT) on mission-critical defense systems, systematically identifying high-risk vulnerabilities and proposing mitigation strategies.
- Authored detailed technical reports with risk analysis, remediation recommendations, and integration pathways for secure communication protocols.
- **Certificate:** [Google Drive PDF](#)

Inland Waterways Authority of India (MoPSW)

Sep 2024 – Oct 2024

Cyber Security Analyst

Noida, India

- Architected cybersecurity infrastructure ensuring data confidentiality, integrity, and availability (CIA triad), aligning with national security standards.
- Developed a Random Forest–based anomaly detection system to identify Advanced Persistent Threats (APTs) and zero-day attacks, integrating ensemble learning for robustness.
- Conducted real-time Network Traffic Analysis (NTA) and system log monitoring, enabling proactive incident response and reducing detection latency.
- **Certificate:** [Google Drive PDF](#)

Projects

AURA: AI Unified Retrieval Assistant | *Python, LangChain, Ollama, Selenium, BeautifulSoup*

Sep 2025

- Designed and implemented an AI-powered information retrieval system capable of extracting, interpreting, and structuring DOM content using natural language queries.
- Integrated Large Language Models (LLMs) with Retrieval-Augmented Generation (RAG) pipelines, enabling context-aware parsing, summarization, and scalable real-time web data extraction.
- Applied advanced prompt engineering and modular architecture for extensibility across diverse web domains.
- **Project Link:** [GitHub Repository](#)

- Developed a mobile emergency response application integrating GPS tracking, geofencing, and real-time route optimization.
- Implemented an AI-driven chatbot for rapid assistance and predictive ML models to optimize dispatch workflows.
- Improved resource allocation and emergency response coordination through data-driven decision-making.
- **Note:** Maintained in a private GitHub repository as part of a startup incubation idea; details available upon request to protect intellectual property.

KISSAN SETU (Smart India Hackathon 2024 Semi-Finalist) | *Flutter, Firebase, OpenAI API, SQLite* Aug 2024

- Built a farmer-to-consumer digital marketplace with dynamic pricing algorithms and an AI-powered chatbot to enhance market efficiency and user engagement.
- Designed secure payment workflows and leveraged agricultural data analytics to deliver actionable insights for policy-making and transparent trade.
- Recognized as a semi-finalist at Smart India Hackathon 2024 for innovation and scalability.
- **Note:** Repository kept private due to ongoing startup development; technical details can be shared in interview discussions.

Additional startup projects are maintained in private repositories to protect intellectual property; details available upon request.

Technical Skills

- **AI/ML Frameworks:** TensorFlow, PyTorch, Hugging Face, Scikit-learn, Keras
- **Generative AI:** Diffusion Models, Variational Autoencoders (VAEs), LangChain, Prompt Engineering, Retrieval-Augmented Generation (RAG)
- **Data Science & Analytics:** Data Mining, Big Data Analytics, Statistical Learning, Predictive Modeling, Feature Engineering
- **Computer Vision & NLP:** OpenCV, Transformers, Reinforcement Learning, Pattern Recognition, Natural Language Processing (NLP)
- **Cybersecurity & Systems:** Information Security, Cryptography, Vulnerability Assessment, Penetration Testing (VAPT), Anomaly Detection
- **Web & Backend Development:** Node.js, Django, REST APIs, SQL/NoSQL Databases, Full-Stack Development
- **Programming Languages:** Python, C++, Java, JavaScript
- **Other:** Technical Writing, Research Documentation, Open Source Mentorship

Certifications

- **Quantum Computing Certification** — CDAC Hyderabad & IIT Roorkee [View Certificate](#)
- **Generative AI Certification** — LangChain & Hugging Face [View Certificate](#)
- **Full-Stack Web Development Certification** — Industry Training Program [View Certificate](#)

Leadership / Extracurricular

Google Developer Group, MAIT (Open Source Mentor)

August 2024 – April 2025

Achievements

- **AI for Humanity 2024 Finalist** — Recognized for impactful, socially relevant AI solutions. [LinkedIn Post](#)
- **Google Cloud Agentic AI Hackathon** — Ranked in the **Top 700 of 9,100+** teams.
- **Best Student Award (2021-2023)** — Awarded by Knowledge 360 Class for academic excellence.
- **Competitive Exams** — Cleared **JEE Advanced 2023** and **GATE 2026**.